$$\int_0^2 \int_0^1 k x (y+x)\, dx\, dy = 1$$

$$\int_0^2 k \left[\int_0^1 xy + x^2\, dx \right] dy = 1$$

$$\int_0^2 k \left[\frac{x^2 y}{2} + \frac{x^3}{3} \right]_0^1 dy = 1$$

$x = 1$

$$\left(\frac{y}{2} + \frac{1}{3} \right) - \left[0 \right]$$

$$\int_1^{10} k \left(\frac{y}{2} + \frac{1}{3} \right) dy$$

$$k \left(\frac{1}{2} \cdot \frac{y^2}{2} + \frac{1}{3} \cdot y \right) = 1$$

$$k \left(\frac{y^2}{4} + \frac{1}{3} y \right) = 1$$

$$k \left(\frac{y^2}{4} + \frac{y}{3} \right)_0^2 = 1$$

$$k \left(\frac{3y^2 + 4y}{12} \right) = 1$$

$$k \left[\left(\frac{1}{2} \cdot \frac{2^2}{2} + \frac{1}{3} \times 2 \right) - \left[0 \right] \right]$$

$$k \left(1 + \frac{2}{3} \right) - 0$$

$$k \left(\frac{5}{3} \right) = 1$$

$$\frac{k}{5/3} = \frac{1}{5/3} \qquad k = \frac{3}{5}$$

$$f(x,y) = \begin{cases} \frac{3}{5} x (y+x) & 0 < x < 1 \\ 0 & 0 < y < 1 \\ & \text{otherwise} \end{cases}$$

$$\int_1^2 \int_0^{1/2} f(x,y)\, dx\, dy$$

$$\int_1^2 \int_0^{1/2} \frac{3}{5} x (y+x)$$

$$\int_1^2 \frac{3}{5} \left(\int_0^{1/2} xy + x^2\, dx \right) dy$$

$$\int_1^2 \frac{3}{5} \left[\frac{x^2 y}{2} + \frac{x^3}{3} \right]_0^{1/2} dy$$

It is more useful when n is small

Example: The PGF of X is

$P(x) = [0.6 + 0.4x]^{10}$

Obtain

i) P_0 (ii) P_2

Soln

P_0 implies $n=0$

$P_0 \Rightarrow P(0) = (0.6 + 0.4(0))^{10}$

$= [0.6]^{10}$

$= 6.04 \times 10^{-3}$

$P_2 \Rightarrow P(2)$ i.e $n=2$

$= [0.6 + 0.4(2)]^{10}$

$= [0.6 + 0.8]^{10}$

$= [1.4]^{10}$

$= 28.93$

Solution

$P_2 = \frac{1}{2!} P''(0)$

$P' = P'(x) = 10 \times 0.4 (0.6 + 0.4x)^9$

$P''(x) = 10 \times 9 \times 0.4 \times 0.4 [0.6 + 0.4x]^8$

$P''(0) = 9 \times 10 \times 0.4 \times 0.4 (0.6)^8$

$P''(0) = 0.242$

$\therefore P_2 = \frac{1}{2!} \times 0.242$

$= 0.121$

Exercise (Study answers)

The random variable X takes the values 0, 1 or 2. The probabilities are $\frac{1}{2}, \frac{1}{4}$ and $\frac{k}{4}$ respectively

Obtain i) PGF of X

ii) The mean and the variance of X from the PGF

Solution

To find k

$P(X = x) = 1$

$\frac{1}{2} + \frac{1}{4} + \frac{k}{4} = 1$

$\frac{2 + 1 + k}{4} = 1$

diagram seems to cluster near a straight line. When the points seems to cluster near a curve, we have a non-linear correlation.

If the points lie close to the line or curve, The relationship is said to be strong. A greater dispersion of the points above the line or curve implies weaker correlation.

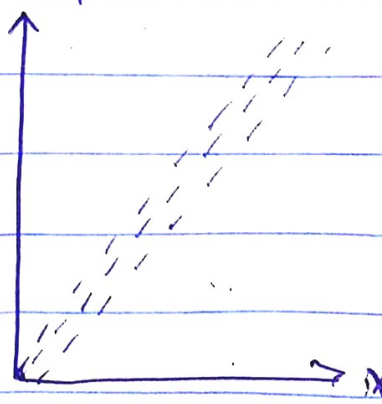
Linear or Non-linear correlation may be positive or negative. We have a positive or direct correlation. If 2 related variable change in the same direction i.e they tend to increase or decrease together. The points on the scattered diagram in this case will be a near curve with positive slope. Should all the points be on the ~~curve~~. Straight line of the curve, we say that correlation is perfect

When the two related variable change in opposite direction i.e X increases as Y decreases and vice versa, we have a negative or linear correlation

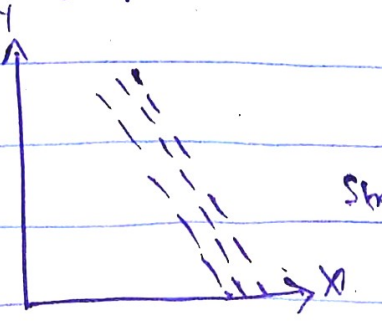
In the case of negative correlation, the points lies near a line or curve with negative slope

Scatter diagrams for the various forms of linear correlation

Examples



Strong positive correlation



Strongly negative correlation

$$= \frac{1}{25}\int_0^{\infty} x\, e^{-(1/5 - t)x}\, dx \quad \cdots (1)$$

$u \quad dv$

$u = x$

$\frac{du}{dx} = 1$

$du = dx$

$dv = e^{-(1/5 - t)x}$

$v = \frac{-5}{1-5t} e^{-(1/5 - t)x}$

$$\int x e^{-(1/5 - t)x} = \left[x \frac{-5}{1-5t} e^{-(1/5 - t)x} \right]_0^{\infty}$$

$$- \int_0^{\infty} \frac{-5}{1-5t} e^{-(1/5 - t)x}\, dx$$

$$\frac{1}{25}\left[\frac{-5}{1-5t} \int_0^{\infty} e^{-(1/5 - t)x}\, dx \right]$$

$$\frac{1}{25}\left[\frac{5}{1-5t} \left[\frac{-5}{1-5t} e^{-(1/5 - t)x} \right]_0^{\infty} \right.$$

$$\frac{1}{25}\left(\frac{5}{1-5t} \left(\frac{5}{1-5t} \right) \right.$$

$$\frac{1}{25} \times \left(\frac{25}{(1-5t)^2} \right. = (1-5t)^{-2}$$

$$\sigma^2 = M_x''(0) - M_x'(0)^2$$

$$= E(x^2) - E(x)^2$$

$$= M_2' - M_1^2$$

$$M_x(t) = (1-5t)^{-2}$$

$$M_x'(t) = -2 \times -5 (1-5t)^{-3}$$

$$= 10 (1-5t)^{-3}$$

$$M_x'(0) = 10 (1-0)^{-3}$$

$$= 10$$

$$M_x''(t) = 10 (1-5t)^{-3}$$

$$= -3 \times 10 \times -5 (1-5t)^{-4}$$

$$= 150 (1-5t)^{-4}$$

$$M_x''(0) = 150 (1-0)^{-4}$$

$$= 150$$

$$\sigma^2 = M_x''(0) - M_x'(0)^2$$

$$= 150 - (10)^2$$

$$= 150 - 100$$

$$= 50$$

Note

$p'(1) = E(x)$

$\text{Var}(x) = E(x^2) - (E(x))^2$

$= p''(1) + p'(1) - (p'(1))^2$

Example: The pgf of x is

$p(x) = [0.3 + 0.7x]^5$

obtain the mean and variance of x

mean $= np$

$= 5 \times 0.7$

$= 3.5$

variance $= np(1-p)$

$= 3.5(0.3)$

$= 1.05$

Soln

$p'(x) = 5 \times 0.7 [0.3 + 0.7x]^{5-1}$

$= 3.5 [0.3 + 0.7x]^4$

$p'(1) = 3.5 (1)^4$

$= 3.5$

$p''(x) = 5 \times 0.7 \times 0.7 \times 4 [0.3 + 0.7x]^3$

$p''(1) = 9.8 [0.3 + 0.7(1)]^3$

$= 9.8 [1]^3$

$= 9.8$

mean $= 3.5$

variance $(x) = p''(1) + p'(1) - (p'(1))^2$

$= 9.8 + 3.5 - (3.5)^2$

$= 9.8 + 3.5 - 12.25$

$= 13.3 - 12.25$

$= 1.05$

Probability from p.g.f

Let $p[x = n] = P_n$, then P_n can be obtained from the pgf as follows.

$P_n = \frac{1}{n!} P_0^{(n)}$ i.e differentiate pgf n times put $x = 0$ and divide the result by $n!$

Note: When n is large, this exercise becomes cumbersome

$$f(1,0) = \frac{\binom{3}{1}\binom{2}{0}\binom{4}{1}}{\binom{9}{2}} = \frac{3 \times 1 \times 4}{36} = \frac{12}{36} = \frac{1}{3}$$

$$f(1,1) = \frac{\binom{3}{1}\binom{2}{1}\binom{4}{0}}{\binom{9}{2}}$$

$$f(0,1) = \frac{\binom{3}{0}\binom{2}{1}\binom{4}{1}}{\binom{9}{2}}$$

$$\frac{3 \times 2 \times 1}{36} = \frac{6}{36} = \frac{1}{6}$$

$$\frac{1 \times 2 \times 4}{36} = \frac{8}{36} = \frac{2}{9}$$

x/y	0	1	2
0	1/6	2/9	1/36
1	1/3	1/6	0
2	1/12	0	0

$$f(2,2) = \frac{\binom{3}{2}\binom{2}{2}\binom{4}{0}}{\binom{9}{2}}$$

$$\frac{\binom{3}{2}\binom{2}{2}\binom{4}{-2}}{\binom{9}{2}}$$

$$\binom{4}{-2} = \frac{4(4-1)(4-2)}{2!}$$

$$= \frac{0 \times 3 \times 2}{36} = 0$$

$$f(0,2) = \frac{\binom{3}{0}\binom{2}{2}\binom{4}{0}}{\binom{9}{2}}$$

$$= \frac{1 \times 1 \times 1}{36} = \frac{1}{36}$$

$$f(1,2) = \frac{\binom{3}{1}\binom{2}{2}\binom{4}{-1}}{\binom{9}{2}} = \frac{3 \times 2 \times 0}{36} = 0$$

$= \frac{2}{3} \times 1 + \frac{2}{3} xy$

$= \frac{4}{3} + \frac{4}{3} y$

$\Rightarrow \frac{1}{3}[1 + 4y]$

$\therefore h(y) = \begin{cases} \frac{1}{3}(1+4y) & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$

Exercise

If the values of the joint probability distribution of X and Y are as shown in the following table

Y \ X	-1	1	$\Rightarrow f(-1)$
-1	$\frac{1}{8}$	$\frac{1}{2}$	$(-1,1)+$ $(-1,0)+$
0	0	$\frac{1}{4}$	$(-1,1)+$ $(1,-1)+$
1	$\frac{1}{8}$	0	$(1,0)+$ $(1,1)$

i) Find the marginal distribution of x and y.

2) If the joint density function of x and y is given by

$f(x,y) = \begin{cases} c(2x+y) & 0<x<1 \\ & 0<y<2 \\ 0 & \text{otherwise} \end{cases}$

i) find the marginal density function of x and y

3) If the random variables x and y have the joint density function

$f(x,y) = \begin{cases} 24xy(1-x-y) & x>0 \\ & y>0 \\ 0 & \text{otherwise} \end{cases}$

Conditional Distribution

Recall that the probability of A given B

$P(A|B) = \frac{P(A \cap B)}{P(B)}$

; $P(B) \neq 0$

ii) $P(x<3, y>0)$

$P(x=1,2,\ y=1,2)$

when $x=1$ & $y=1$

$\Rightarrow 0.1$

when $x=1$ and $y=2$

$\Rightarrow 0.02$

when $x=2$ and $y=1$

$\Rightarrow 0.16$

when $x=2$ and $y=2$

$\Rightarrow 0.2$

$P(0.1+0.02+0.16+0.2)$

$= 0.48$

Definition: If X and Y are discrete random variables, the function given by $F(x,y) \Rightarrow$

$P(X \leq x, Y \leq y) = \sum_{s \leq x} \sum_{t \leq y} f(s,t)$

$-\infty < x < \infty$

$-\infty < y < \infty$

where $f(s,t)$ is the value of the joint probability distribution of X and Y at (s,t) is called the joint distribution function or joint cumulative distribution of X and Y

Theorem: A bivariate function can serve as a joint probability density function of a pair of continuous random variables X and Y. If its values $f(x,y)$ satisfies the conditions.

1) $f(x,y) \geq 0$ for $-\infty < x < \infty$, $-\infty < y < \infty$

2) $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y)\, dx\, dy = 1$

Example: If the joint density function of X and Y is given by

$$f(x,y) = \begin{cases} kx(y+x) & 0<x<1 \\ & 0<y<2 \\ 0 & \text{otherwise} \end{cases}$$

find $P(x,y) \in A$, where A is the region $\{(x,y): 0<x<\frac{1}{2},\ 1<y<2\}$

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Probability Generating function

The probability Generating function is one which literally generates probability of an integer valued random variable X. let X, the probability mass function

X (pmf) $\{P_n, n \geq 0\}$

i.e $P[X=n] = P_n$

$n = 0, 1, 2, \ldots$

The probability generating function (PGF) of X written $P(X)$ is given by

$$P(X) = \sum_{n \geq 0} P_n X^n$$

$$= P_0 + P_1 X + P_2 X^2 + P_3 X^3 + \ldots$$

Note! The coefficient of the polynomials are the nth probabilities

Example: Given the distribution

X	0	1	2	3
P_n	0.1	0.3	0.4	0.2

$$P(X) = P_0 + P_1 X + P_2 X^2 + P_3 X^3$$

$$= 0.1 + 0.3X + 0.4X^2 + 0.2X^3$$

Note $P(1) = 1$, when $\{P_n \; n \geq 0\}$ is a distribution, i.e the PGF when evaluated as $X = 1$ is 1

2) $P'(1) = E(X)$ when differentiated once and evaluated at $X = 1$, the result is expected of X $(E(X))$ and that is the mean.

$$P(X) = \sum_{n \geq 0} P_n X^n$$

$$\frac{dP(X)}{dX} = P'(X) = \sum_{n \geq 0} n P_n X^{n-1}$$

$$P'(1) = \sum_{n \geq 0} n P_n = E(X)$$

Suppose A and B are the events $X=x$ and $Y=y$, then

$p(X=x|Y=y)$

$= p(X=x, Y=y) = \frac{f(x,y)}{h(y)}$

Definition: If $f(x,y)$ is the value of the joint probability distribution of the discrete random variables X and Y at (x,y) and $h(y)$ is the value of the marginal distribution of Y at y, the function given by

$$f(x|y) = \frac{f(x,y)}{h(y)}$$

$h(y) \neq 0$ is called the conditional distribution of X given $Y=y$.

Correspondingly, the function given by

$$w(y|x) = \frac{f(x,y)}{g(x)} \quad g(x) \neq 0$$

is called the conditional distribution of Y given $X=x$

Example: Consider the table

X \ Y	0	1	2
0	1/6	1/3	1/2
1	2/9	1/6	
2	1/36		

Find the conditional distribution of X given $Y=1$

$$F(x|Y=1) = \frac{f(x, y=1)}{h(y=1)}$$

$f(0,1) \Rightarrow 2/9$

$h(y=1) : (0,1) + (1,1)$

$$\frac{2}{9} + \frac{1}{6} \quad \frac{4+3}{18} = \frac{7}{18}$$

$$f(a|1) = \frac{\frac{2}{9}}{\frac{7}{18}} \Rightarrow \frac{2}{9} \times \frac{18}{7}^2 = \frac{4}{7}$$

$$M_x(t) = E(e^{tx}) = \sum_x e^{tx} \cdot f(x)$$

When x is discrete

$$M_x(t) = E(e^{tx}) = \int_{-\infty}^{\infty} e^{tx} \cdot f(x)\, dx$$

when x is continuous

Note) $e^{tx} = 1 + tx + \frac{t^2x^2}{2!} + \frac{t^3x^3}{3!} + \cdots + \frac{t^r x^r}{r!}$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^r}{r!}$$

Example

Given that X has the probability distribution

$$f(x) = \begin{cases} \frac{1}{8}\binom{3}{x} & \text{for } x = 0,1,2,3 \\ 0 & \text{otherwise} \end{cases}$$

Find the moment generating function of the random variable and use it to determine μ_1', μ_2' and σ

Soln

$$M_x(t) = E(e^{tx}) = \sum_x e^{tx} \cdot f$$

$$= 3 \sum_{x=0} e^{tx} \cdot \frac{1}{8}\binom{3}{x}$$

$$\frac{1}{8} \sum_{x=0}^{3} e^{tx} \cdot \binom{3}{x}$$

when $x = 0$

$$\frac{1}{8}\left[e^{t(0)} \cdot \binom{3}{0}\right]$$

$$\frac{1}{8}(1 \times 1) = 1$$

when $x = 1$

$$\left(e^{t(1)} \cdot \binom{3}{1}\right]$$

$$e^t \times 3 = 3e^t$$

when $x = 2$

$$e^{t(2)} \cdot \binom{3}{2}$$

$$e^{2t} \cdot 3 = 3e^{2t}$$

$g(x) = \int_{-\infty}^{\infty} f(x,y)\,dy \qquad -\infty < x < \infty$

is called the marginal density of X

correspondingly, the function given by $h(y) = \int_{-\infty}^{\infty} f(x,y)\,dx$

$-\infty < y < \infty$

Example :-

Given the Joint density $f(x,y)$

$$= \begin{cases} \frac{2}{3}(x+2y) & 0<x<1 \\ & 0<y<1 \\ 0 & \text{otherwise} \end{cases}$$

Find the marginal of x and y

Sol:-

$g(x) \int_{-\infty}^{\infty} f(x,y)\,dy$

$\int_0^1 \frac{2}{3}(x+2y)\,dy$

$\frac{2}{3}\int_0^1 (x+2y)\,dy$

$\frac{2}{3}(xy + y^2)\Big|_0^1$

$\frac{2}{3}\left[x(1) + (1)^2\right]$

$\frac{2}{3}(x+1)$

$g(x) = \begin{cases} \frac{2}{3}(x+1) & 0<x<1 \\ 0 & \text{otherwise} \end{cases}$

$h(y) = \int_{-\infty}^{\infty} f(x,y)\,dx$

$\int_0^1 \frac{2}{3}(x+2y)\,dx$

$\frac{2}{3}\left[\frac{x^2}{2} + 2xy\right]_0^1$

$\frac{2}{3}\left[\frac{1}{2} + 2y\right]$

$h(y) = \begin{cases} \frac{2}{3}\left(\frac{1}{2}+2y\right) & 0<y<1 \\ 0 & \text{otherwise} \end{cases}$

Example: X and Y are jointly distributed with $f(x,y)$

$$f(x,y) = \begin{cases} e^{-x+y} & x>0, y>0 \\ 0 & \text{otherwise} \end{cases}$$

i) Calculate $P(X \le 1, Y \ge 1)$

ii) Calculate $P(1 \le X \le 2 \mid Y \le 2)$

Solution

$P(X \le 1, Y \ge 1)$

$x = (0, 1)$

$y = (1, \infty)$

$$\int_1^{\infty} \int_0^{1} e^{-(x+y)} \, dx \, dy$$

$$\int_1^{\infty} \int_0^{1} e^{-x} \cdot e^{-y} \, dx \, dy$$

$$\int_1^{\infty} e^{-y} \left[\int_0^{1} e^{-x} \, dx \right] dy$$

$$\int_1^{\infty} e^{-y} \left(-e^{-x} \right)_0^1 \, dy$$

$$\int_1^{\infty} e^{-y} \left(-e^{-1} - (-e^{0}) \right) dy$$

$$\int_1^{\infty} e^{-y} \left[-e^{-1} + 1 \right] dy$$

$$\int_1^{\infty} -e^{-1} \cdot e^{-y} + e^{-y} \, dy$$

$$\left[-e^{-1} - e^{-y} + (-e^{-y}) \right]_1^{\infty}$$

$$-e^{-1} \quad -e^{-y} \quad -e^{-y}$$

$$\left[-e^{-1} - 2e^{-y} \right]_1^{\infty}$$

$$\left[-e^{-1} - 2e^{-\infty} \right] - \left[-e^{-1} - 2e^{-1} \right]$$

$$\left[-e^{-1} - 2e^{-\infty} \right] + \left[-3e^{-1} \right]$$

$$\left(e^{-(1+y)} - e^{-y} \right)_1^{\infty}$$

$$\left(e^{-y} + e^{-(1+y)} \right)_1^{\infty}$$

3) $F(-\infty, -\infty) = P(x \leq -\infty, y \leq -\infty) = 0$ probability of an empty or null set

4) $f(x, -\infty) = F(-\infty, y) =$ joint distribution of x with a null set.

Example 1 If (x,y) are jointly distributed & $f(x,y) = k(6-x-y)$ $0 < x < 2$, $2 < y < 4$

obtain

i) the value of k

ii) the probability of ie $P(x<1, y<3)$

i) $P(x > 1/2, y < 3 1/2)$

solution

$$\int_2^4 \int_0^2 k(6-x-y)\,dx\,dy = 1$$

$$\int_2^4 k\left[\int_0^2 (6-x-y)\,dx\right] dy = 1$$

$$\int_2^4 k\left[6x - \frac{x^2}{2} - xy\right]_0^2 dy = 1$$

when $x = 2$

$$6(2) - \frac{2^2}{2} - 2y$$

$$12 - 2 - 2y$$

$$10 - 2y$$

$$\int_2^4 k(10-2y)\,dy = 1$$

$$k\left[10y - \frac{2y^2}{2}\right]_2^4 = 1$$

$$k\left[10y - y^2\right]$$

when $y = 4$

$$k([40-16] - [20-4]) = 1$$

$$k(24-16) = 1$$

$$k(8) = 1$$

$$8k = 1$$

$$k = 1/8$$

Theorem: A bivariate function can serve as the joint probability distribution of a pair of discrete random variables X and Y, if and only if its values $f(x,y)$ satisfies the conditions

1) $f(x,y) \geq 0$ for each pair of values (x,y) within its domain

2) $\sum_x \sum_y f(x,y) = 1$, where the double summation extends over all possible pairs (x,y) within its domain.

Example:

Determine the value of k so that the function $f(x,y) = kxy$ for $x = 1,2,3$, $y = 1,2,3$ can serve as a joint probability distribution.

Solution

$f(x,y) = kxy$

$x = 1,2,3$

$y = 1,2,3$

$f(x,y) = kxy$

$f(1,1) = k \times 1 \times 1$

$= k$

$f(1,2) = k \times 1 \times 2$

$= 2k$

$f(1,3) = k \times 1 \times 3$

$= 3k$

$f(2,1) = k \times 2 \times 1$

$= 2k$

$f(2,2) = k \times 2 \times 2$

$= 4k$

$f(2,3) = k \times 2 \times 3$

$= 6k$

$f(3,1) = k \times 3 \times 1$

$= 3k$

$f(3,2) = k \times 3 \times 2 = 6k$

$f(3,3) = k \times 3 \times 3 = 9k$

$\sum_x \sum_y f(x,y) = 36k$

$36k = 1$

$k = \frac{1}{36}$

$$\int_2^4 \int_0^2 \frac{1}{8}(6-x-y)\,dx\,dy = 1$$

$$\int_2^3 \frac{1}{8}\left(\frac{11-2y}{2}\right)dy$$

ii) $P(X<1, Y<3)$

$x = 0, 1$

$y = 2, 3$

$$\int_2^3 \int_0^1 \frac{1}{8}(6-x-y)\,dx\,dy$$

$$\int_2^3 \frac{1}{8}\left[\int_0^1 6-x-y\,dx\right]dy$$

$$\int_2^3 \frac{1}{8}\left[6x - \frac{x^2}{2} - xy\right]_0^1 dy$$

$$\int_2^3 \frac{1}{8}\left[6 - \frac{1}{2} - y\right]dy$$

$$\int_2^3 \frac{1}{8}\left(\frac{11}{2} - \frac{y}{2}\right)dy$$

$$\frac{1}{8}\int_2^3 \left(\frac{12-1-2y}{2}\right)dy$$

$$\frac{1}{8}\cdot\frac{1}{2}\int_2^3 11-2y\,dy$$

$$\frac{1}{16}\left[11y - \frac{2y^2}{2}\right]_2^3$$

$$\frac{1}{16}\left[(11(3) - (3)^2) - (11(2) - (2)^2)\right]$$

$$\frac{1}{16}\left[(33-9) - (22-4)\right]$$

$$\frac{1}{16}\left[24 - 18\right]$$

$$\frac{1}{16}(6) = \frac{6}{16} = \frac{3}{8}$$

iii) $P(x > 1/2, y < 3/2)$

$$\int_2^{3/2} \int_{1/2}^2 \frac{1}{8}(6-x-y)\,dx\,dy$$

$= C(x^2+y^2)$

for $x = -1, 0, 1, 3$

and $y = -1, 2, 3$.

i) find the value of C

ii) $P(x=0, Y \leq 2)$

iii) $P(x \leq 1, y \geq 2)$

iv) $P(X > 2-y)$

3- Determine k so that

$$f(x,y) = \begin{cases} kx(x-y) & \text{for } 0 < x < 1 \\ & -x < y < x \\ 0 & \text{Otherwise} \end{cases}$$

can serve as a probability density function

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MARGINAL DISTRIBUTION

Definition

If X and Y are discrete random variable and $f(x,y)$ is the value of their joint probability distribution at (x,y) the function $g(x) = \sum_y f(x,y)$ for each x within the range of X is called the marginal distribution of X.

Correspondingly, the function given by $h(y) = \sum_x f(x,y)$ for each y within the range of Y is called the marginal distribution of Y.

$\Rightarrow$

If x and Y are continuous random variables and $f(x,y)$ is the value of their joint probability density at (x,y) the function given by

when $x = 3$

$$e^{t(3)} \cdot \binom{3}{3}$$

$$e^{3t} \times 1 = e^{3t}$$

$$\Rightarrow \frac{1}{8}\left[1 + 3e^{t} + 3e^{2t} + e^{3t}\right]$$

1

1 1

1 2 1

1 3 3 1

1 4 6 4 1

$$\frac{1}{8}\left[1 + e^{t}\right]^{3}$$

$$\downarrow$$

$$(1+e^{t})(1+e^{t})$$

$$(1 + e^{t} + e^{t} + e^{2t})$$

$$(1 + 2e^{t} + e^{2t})(1+e^{t})$$

$$\Rightarrow 1 + e^{t} + 2e^{t} + 2e^{2t} + e^{2t} + e^{3t}$$

$$\Rightarrow 1 + 3e^{t} + 3e^{2t} + e^{3t} \;//$$

So the MGF $= \frac{1}{8}(1+e^{t})^{3}$

$$M_1' = \frac{d\,M_x(t)}{dt}\Big|_{t=0}$$

$$\frac{1}{8}(1+e^{t})^{3}$$

$$\frac{1}{8} \times 3 \times e^{t}(1+e^{t})^{3-1}$$

$$\frac{3}{8} e^{t}(1+e^{t})^{2}$$

at $t=0$

$$\frac{3}{8} e^{(0)}(1+e^{0})^{2}$$

$$\frac{3}{8}(2)^{2} \Rightarrow \frac{3}{8} \times 4$$

$$= \frac{12}{8} \Rightarrow \frac{3}{2}$$

Note that $M_1' =$ mean $\Rightarrow E(x)$

$$M_2' = \frac{d^2(M_x(t))}{dt^2}\Big|_{t=0}$$

$$= M_x''(0)$$

$$m_x' = \frac{3}{8}(1+e^{t})^{2} e^{t}$$

$$= \frac{3}{8} e^{t}(1+e^{t})^{2}$$

We have a strong negative correlation. When r approaches zero (0), the correlation is weak. A value of r=0 indicates absence of correlation.

The Spearman rank correlation coefficient is given by

$$r = \frac{n\Sigma xy - (\Sigma x)(\Sigma y)}{\sqrt{n\Sigma x^2 - (\Sigma x)^2 \cdot n\Sigma y^2 - (\Sigma y)^2}}$$

x	y	xy	x^2	y^2
1.5	1.0	1.5	2.25	1
2.0	1.5	3	4	2.25
2.5	1.8	4.5	6.25	3.25
3.0	2.2	6.6	9.0	4.84
3.5	2.6	9.1	12.25	6.76
4.0	3.0	12.0	16.0	9.0
4.5	3.3	14.85	20.25	10.89
5.0	3.8	19.0	25.00	14.44
5.5	4.0	22	30.25	16.00
6.0	4.0	24.0	36.0	16.00
37.5	27.2	161.55	16.25	84.08

Example

Given the data below on annual family income and expenditure of a sample of 10 families

Annual family income (x) (000, naira)	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0	5.5	6.0
Expenditure (y)	1.0	1.5	1.8	2.2	2.6	3.0	3.3	3.8	4.0	4.0

Calculate and interpret the coefficient of correlation

This result indicate there is a strong positive relation between Annual income (x) and Annual

3) $P(0) = P[X=0] = P_0$

Example 4

A random variable X has the binomial distribution which is $b(n,p)$ obtain the pgf of X.

Soln

$$P(x) = \sum_r P_r X^r$$

$X \sim b(n,p) \Rightarrow$

$${}^nC_r\, p^r q^{n-r}$$

$$\Rightarrow P(x) = \sum_r {}^nC_r\, p^r q^{n-r} X^r$$

$$P(x) = \sum_{r=0}^{n} {}^nC_r (Px)^r \cdot q^{n-r}$$

Note

$$\sum_{r=0}^{n} {}^nC_r\, a^r b^{n-r} = (a+b)^n$$

So $= [q + P(x)]^n$

Example : A random variable X has the poisson distribution

$$P(X=r) \quad \frac{\lambda^r e^{-\lambda}}{r!}$$

$; r = 0, 1, \ldots$

obtain the PGF of X

$$P(x) = \sum_r P_r X^r$$

$X \sim P(\lambda) \Rightarrow$

$$P_r = \frac{\lambda^r e^{-\lambda}}{r!}$$

$$\sum_{r=0}^{\infty} \frac{\lambda^r e^{-\lambda}}{r!} \cdot X^r$$

$$\sum_{r=0} \frac{(\lambda x)^r e^{-\lambda}}{r!}$$

$$= e^{-\lambda} \cdot e^{\lambda x}$$

$$= e^{-\lambda(1-x)}$$

Note

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} \ldots$$

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Course Outline

1) Joint probability distribution
2) Marginal distribution
3) conditional probability distribution
4) probability Generating function (P.G.F)
5) Moment Generating function (M.G.F)
6) Covariance and correlation
7) Change of Variables
8) probability distribution of function of random variables

$3 + k = 4$

$k = 4 - 3$

$k = 1$

x	0	1	2	3
$P(x=x)$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{4}$	

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Moments | Moment Generating functions

Moments: The Rth moment about the origin of a random variable X denoted by M_r' is the expected value of X^r, symbolically

$$M_r = E(X^r) = \sum_x x^r f(x)$$

for $r = 0, 1, 2, 3 \cdots$ when x is discrete, and

$$M_r' = E(x^r) = \int_{-\infty}^{\infty} x^r \cdot f(x)\, dx$$

when x is continuous

M_2 is called the variance of the distribution of X or simply variance of X and is denoted by σ^2, $\text{var}(x)$ or $\text{Var}(x)$

σ, the positive square root of the variance is called the standard deviation.

Theorem $\sigma^2 = M_2' - M_1^2$

Proof:

$$\sigma^2 = E[(x - M)^2]$$

$$= E[(x - M)(x - M)]$$

$$= E(x^2 - Mx - Mx + M^2)$$

$$= E(x^2 + 2Mx + M^2)$$

$$\Rightarrow E(x^2) - 2M E(x) + M^2$$

$$\Rightarrow E(x^2) - 2M \cdot M + M^2$$

$$\Rightarrow E(x^2) - 2M^2 + M^2$$

$$E(x^2) - M^2$$

$$= M_2' - M_1^2$$

Definition: The moment Generating function of a random variable X, where it exist is given by

Example: Given the function

$$f(x) = \begin{cases} k x e^{-x/5} & x > 0 \\ 0 & o.w \end{cases}$$

i) find $M_x(t)$, hence obtain σ

Solution

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

$$\int_{0}^{\infty} k\,x e^{-x/5}\,dx = 1$$

$$k\int_{0}^{\infty} x e^{-x/5}\,dx = 1$$

$u \quad dv$

$u = x$

$\frac{du}{dx} = 1 \quad du = dx$

$dv = e^{-x/5} \quad v = -5e^{-x/5}$

$\int u\,dv = uv - \int v\,du$

$= -5x e^{-x/5} - \int_0^{\infty} -5e^{-x/5} = 1$

$= \left[-5x e^{-x/5}\right]_0^{\infty} - \int_0^{\infty} -5e^{-x/5} = 1$

$= \left[-5x e^{-x/5}\right]_0^{\infty} - \int_0^{\infty} -5e^{-x/5} = 1$

$k\left[0 - (-5)\int e^{-x/5}\right] = 1$

$k\left(5 \times 5 e^{-x/5}\right) = 1$

$k\left[-25\left[e^{-x/5}\right]_0^{\infty}\right] = 1$

$k\left[-25\left[e^{-\infty/5}\right] - \left[e^{-0/5}\right]\right] = 1$

$k(-25\ (0-1)) = 1$

$k(-25(-1)) = 1$

$k(25) = 1$

$k = \frac{1}{25}$

$$f(x) = \begin{cases} \frac{1}{25} x e^{-x/5} & x > 0 \\ 0 & \text{otherwise} \end{cases}$$

$$\int_0^{\infty} \frac{1}{25} x e^{-x/5}\,dx$$

$$f(1,1) = \frac{1}{\frac{6}{7}} \Rightarrow \frac{1}{6} \times \frac{7^3}{7} = \frac{3}{7}$$

$$f(2,1) = \frac{0}{\frac{7}{18}} = 0$$

$$f(x|y=1) \Rightarrow \begin{cases} 4/7 & \text{for } x=0, y=1 \\ 3/7 & \text{for } x=1, y=1 \\ 0 & \text{Otherwise} \end{cases}$$

For continuous random variables,

$$f(x|y) = \frac{f(x,y)}{h(y)} \text{ where}$$

$$h(y) = \int_{-\infty}^{\infty} f(x,y)\,dx$$

This is called the conditional density of X given $Y=y$

correspondingly, the conditional density of Y given $X=x$ is given by

$$w(y|x) = \frac{f(x,y)}{g(x)}$$

$$g(x) = \int_{-\infty}^{\infty} f(x,y)\,dy$$

Example:

Given the joint density function

$$f(x,y) = \begin{cases} 4xy & 0<x<1 \\ & 0<y<1 \\ 0 & \text{Otherwise} \end{cases}$$

1) Find the marginal density of X and Y and conditional density of X given $Y=y$ and

$$g(x) = \int_0^1 4xy\,dy$$

$$= \left[\frac{4xy^2}{2}\right]_0^1$$

$$= \left[2xy^2\right]_0^1$$

$$2x(1)^2 = 2x$$

$$g(x) = \begin{cases} 2x & 0<x<1 \end{cases}$$

$h(y) = \int_0^1 f(x,y)\,dx$

$= \int_0^1 4xy\,dx$

$\left[\frac{4x^2y}{2}\right]_0^1$

$\left[2x^2y\right]_0^1$

$2(1)^2 y = 2y$

$h(y) = \begin{cases} 2y & 0<y<1 \\ 0 & \text{otherwise} \end{cases}$

Conditional distribution of X

$Y=y$

$f(x|Y=y) = \frac{f(x, Y=y)}{h(y)}$

$= \frac{4xy}{2y} = 2x$

Exercises

1) If the joint density function of X and Y is given by

$f(x,y) = \begin{cases} \frac{1}{4}(2x+y) & 0<x<1 \\ & 0<y<2 \\ 0 & \text{otherwise} \end{cases}$

find (i) the marginal density of X

ii) The marginal density of Y

iii) the Conditional density of X given $X=1$

iv) The condition density of Y given $X = 1/2$

$-e^{-\infty} + e^{-(1+\infty)} - (-e^{-1} + e^{-2})$

$\Rightarrow 0 + e^{-1} - e^{-2}$

$= 0.368 - 0.135$

$= 0.233$

2) $p(1 \le x \le 2, y \le 2)$

$x = 1, 2)$

$y = (0, 1, 2)$

$$\int_0^2 \int_1^2 e^{-(x+y)} \, dx \, dy$$

$$\int_0^2 \int_1^2 e^{-x} e^{-y} \, dx \, dy$$

$$\int_0^2 e^{-y} \left[-e^{-x} \right]_1^2 dy$$

$$\int_0^2 e^{-y} \left[(e^{-2}) - (-e^{-1}) \right] dy$$

$$\int_0^2 e^{-y} \left[-e^{-2} + e^{-1} \right] dy$$

$$\int_0^2 e^{-y} (0.135 + 0.233) \, dy$$

$$\int_0^2 e^{-y} (0.233) \, dy$$

$$-e^{-y} (0.233)$$

$$= \left[-0.233 \cdot e^{-y} \right]_0^2$$

$$\left[-0.233 e^{-2} \right] - \left[-0.233 \times e^{0} \right]$$

$$-0.03153 + 0.233$$

$$= 0.202 //$$

1) If x and Y have the Joint probably

y | x	0	1	2
0	1/12	1/16	1/24
1	1/4	1/4	1/40
2	1/5	1/20	–
3	1/120	–	–

Find i) $p(x = 1, y = 2)$

ii) $p(x = 0, 1 \le y \le 3)$

iii) $p(X + Y \le 1)$

iv) $p(X > Y)$

2) If the Joint probability distribution x and Y is given by $f(x, y)$

If $r = 1/36$

$f(1,1) = 1/36$ $\quad f(1,2) = 2/36$

$f(1,3) = 3/36$ $\quad f(2,1) = 2/36$

$f(2,2) = 4/36$ $\quad f(2,2) = 5/36$

$f(3,1) = 3/36$ $\quad f(3,2) = 6/36$

$f(3,3) = 9/36$

Summing all $= \frac{36}{36} \Rightarrow 1$

it satisfies the conditions so it can serve as Joint probability distribution.

STA 2.1f $\quad$ 20/05/26

$f(x,y) =$

x/y	0	1	2
1	c	0.1	0.02
2	0.06	2c	0.2
3	0.11	0.21	0.06

i) Obtain the value of c

ii) $p(X \leq 2, y > 1)$

iii) $p(x < 3, y > 0)$

Solution

note $\sum f(x,y) = 1$

$c + 0.1 + 0.02 + 0.06 + 2c + 0.2 + 0.11 + 0.21 + 0.06$

$3c + 0.76 = 1$

$3c = 1 - 0.76$

$\frac{3c}{3} = \frac{0.24}{3}$ $\quad c = 0.08$

x/y	0	1	2
1	0.08	0.1	0.02
2	0.06	0.16	0.2
3	0.11	0.21	0.6

ii) $p(X \leq 2, y > 1)$

$p(x = 1,2, y = 2)$

when $x = 1$ and $y = 2$

$= 0.02$

when $x = 2$ and $y = 2$

$= 0.2$

$p(x \leq 2, y > 1)$

$(0.02 + 0.2) \Rightarrow 0.22$

(ii) $P(X<3;\ Y>0)$

$P(x=1,2;\ y=1,2)$

when $x=1$ & $y=1$

$\Rightarrow 0.1$

when $x=1$ and $y=2$

$\Rightarrow 0.02$

when $x=2$ and $y=1$

$\Rightarrow 0.16$

when $x=2$ and $y=2$

$\Rightarrow 0.2$

$P(0.1+0.02+0.16+0.2)$

$= 0.48$

Definition: If X and Y are discrete random variables, the function given by $F(x,y) \Rightarrow$

$P(X \geq x, Y \leq y) = \sum_{s \leq x} \sum_{t \leq y} f(s,t)$

$-\infty < x < \infty$

$-\infty < y < \infty$

where $f(s,t)$ is the value of the joint probability distribution of X and Y at (s,t) is called the joint distribution function or joint cumulative distribution of X and Y

Theorem: A bivariate function can serve as a joint probability density function of a pair of continuous random variables X and Y. If its values $f(x,y)$ satisfies the conditions.

1) $f(x,y) \geq 0$ for $-\infty < x < \infty$, $-\infty < y < \infty$

2) $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y)\, dx\, dy = 1$

Example: If the joint density function of X and Y is given by $f(x,y) = \begin{cases} x(1+y) & 0<x<1,\ 0<y<2 \\ 0 & \text{otherwise} \end{cases}$

find $P(x,y) \in A$, where A is the region $\{(x,y): 0<x<\frac{1}{2},\ 1<y<2\}$

$$\frac{1}{25}\int_0^\infty x e^{-x/5}$$

u $\quad$ dv

$u = x$

$\frac{du}{dx} = 1 \quad du = dx$

$dv = e^{-x/5} \quad v = -5e^{-x/5}$

$$\int u\,dv = uv - \int v\,du$$

$$= x(-5e^{-x/5}) - \int -5e^{-x/5}\,dx$$

$$= \frac{1}{25}\left[-5xe^{-x/5}\right]_0^\infty - (-5)\int_0^\infty e^{-x/5}\,dx$$

$$= \frac{1}{25}\left[5\int_0^\infty e^{-x/5}\,dx\right]$$

$$\frac{1}{25}\left[-25\left[e^{-x/5}\right]_0^\infty\right]$$

$$\int x e^{-x/5} = x(-5e^{-x/5}) - \int -5e^{-x/5}\,dx$$

$$= -5xe^{-x/5} + 5\int e^{-x/5}\,dx$$

$$= -5xe^{-x/5} + 5(-5e^{-x/5})$$

$$\int$$

$$= -5xe^{-x/5} - 25e^{-x/5}$$

$$\int_0^\infty x e^{-x/5}\,dx = \left[-5xe^{-x/5} - 25e^{-x/5}\right]$$

and $x \to \infty$

$e^{-x/5} = 0$

$-5xe^{-x/5} - 25e^{-x/5} = 0$

at $x = 0$

$-5(0)e^0 - 25e^0 = -25$

$= 0 - (-25)$

$= 25$

Solving:

$M_x(t) = E(e^{tx})$

$$= \int_{-\infty}^{\infty} e^{tx} \cdot f(x)\,dx$$

$$= \int_0^\infty e^{tx} \cdot \frac{1}{25} x e^{-x/5}\,dx$$

$$= \frac{1}{25}\int_0^\infty e^{tx} \cdot x e^{-x/5}$$

$$\frac{3}{5}\left[\frac{(\frac{1}{2})^2 y}{2} + \frac{(\frac{1}{2})^3}{3}\right] - 0$$

$$\left(\frac{\frac{1}{4}y}{2} + \frac{\frac{1}{8}}{3}\right)$$

$$\left(\frac{\frac{y}{4}}{2} + \frac{\frac{1}{8}}{3}\right) \Rightarrow \left[\frac{y}{8} + \frac{1}{24}\right]$$

$$\frac{3}{5}\left[\frac{1}{8}\cdot\frac{y^2}{2} + \frac{1}{24}\cdot y\right]_1^2$$

$$\frac{3}{5}\left[\left(\frac{4}{16} + \frac{2}{24}\right) - \left(\frac{1}{16} + \frac{1}{24}\right)\right]$$

$$\frac{3}{5}\left(\frac{1}{3} - \frac{5}{48}\right)$$

$$\frac{3}{5}\cdot\left(\frac{11}{48}\right) = \frac{11}{80}$$

STA 214 3/06/2021

Definition

If X and Y are continuous random variables, the function given by

$$f(x,y) = P(X \le x, Y \le y)$$

$$= \int_{-\infty}^{y}\int_{-\infty}^{x} f(s,t)\,ds\,dt$$

$-\infty < x < \infty$

$-\infty < y < \infty$

where $f(s,t)$ is the value of the joint probability density of x and y at (s,t) is called joint ~~the~~ distribution of x and Y

NOTE: $f(x,y) = \frac{\delta^2}{\delta x \delta y} f(x,y)$

Some properties of the distribution function $f(x,y)$

i) $0 \le f(x,y) \le 1$

ii) $f(\infty,\infty) = P(x \le \infty, y \le \infty) = 1$

which is the probability of the entire sample space

$u \quad v$

$$\mu_2' = \frac{3}{8}(1+e^t)^2 \cdot e^t$$

$u = (1+e^t)^2$

$\frac{du}{dt} = e^t \times 2(1+e^t)$

$\frac{du}{dt} = 2(1+e^t) \cdot e^t$

$v = e^t$

$\frac{dv}{dt} = e^t$

$$\frac{3}{8}\left[v\frac{du}{dt} + u\frac{dv}{dt}\right]$$

$$\frac{3}{8}\left[e^t \cdot [2(1+e^t)\cdot e^t] + (1+e^t)^2 (e^t)\right] \text{ at } t=0$$

$$\frac{3}{8}\left[e^0 \cdot [2(1+e^0)\cdot e^0] + [1+e^0)^2 \cdot (e^0)\right]$$

$$\frac{3}{8}[1 \times 2(1+1)\cdot 1] + (1+1)^2 \cdot 1)$$

$$\frac{3}{8}(1\times 4) + 4)$$

$$\frac{3}{8}(8) = 3$$

$$\mu_2' = 3$$

$\sigma^2 = 3$

$\therefore \sigma = \sqrt{3}$

$\therefore \sigma^2 = 3$

$\sqrt{\sigma^2} = \sqrt{3}$

$\sigma = \sqrt{3}$

$$\sigma^2 = \mu_2' - \mu_1'$$

$$\sigma^2 = 3 - \left(\frac{3}{2}\right)^2$$

$$= \frac{3}{1} - \frac{9}{4}$$

$$\frac{12-9}{4} = \frac{3}{4}$$

$$\sigma^2 = \frac{3}{4}$$

$$\sigma = \sqrt{\frac{3}{4}} \qquad \sigma = \frac{\sqrt{3}}{2}$$

$$\sigma = 0.866$$

Joint probability Distribution

Definition If X and Y are discrete random variables the function given by $f(x,y) = P(X=x, Y=y)$ for each pair of values (x,y) within the range of X and Y is called the Joint probability function or Joint probability distribution of X and Y

Example:

Two tablet are selected at random from a bottle containg three Aspirin, 2 sedative and 4 Laxative tablets. If X and Y are respectively the number of Aspirin tablet and the number of Sedative tablet included among the two tablets drawn from the bottle. Find the probability associated with all possible value (x,y)

The range of $x = 0, 1, 2$

The range of $y = 0, 1, 2$

Solution

X: aspirin = 3

Y: sedative = 2

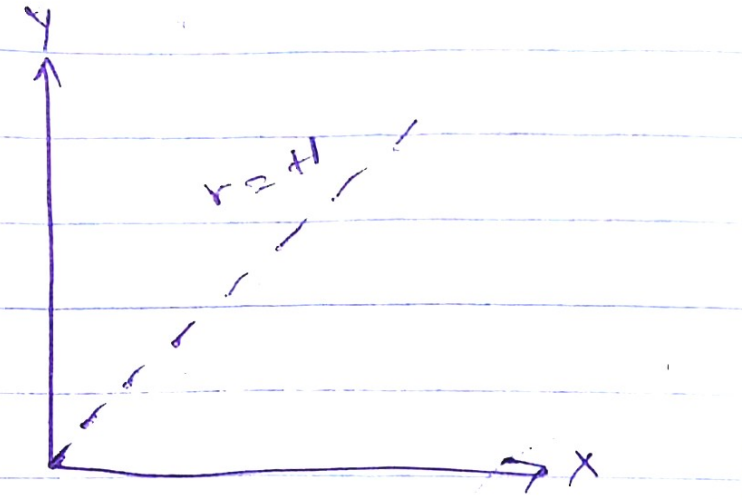
Total = 3 + 2 + 4 = 9

$$f(x,y) = \frac{\binom{3}{x}\binom{2}{y}\binom{4}{2-x-y}}{\binom{9}{2}}$$

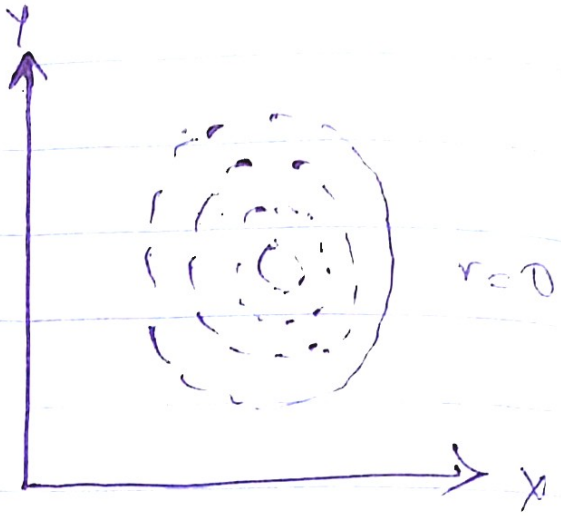
$$f(0,0) = \frac{\binom{3}{0}\binom{2}{0}\binom{4}{2}}{\binom{9}{2}}$$

$$= \frac{1 \times 1 \times 6}{36} = \frac{6}{36}$$

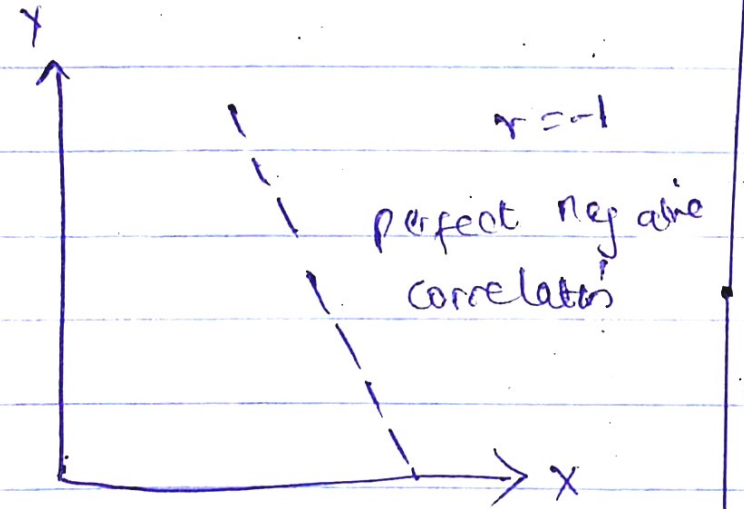
$$= \frac{1}{6}$$

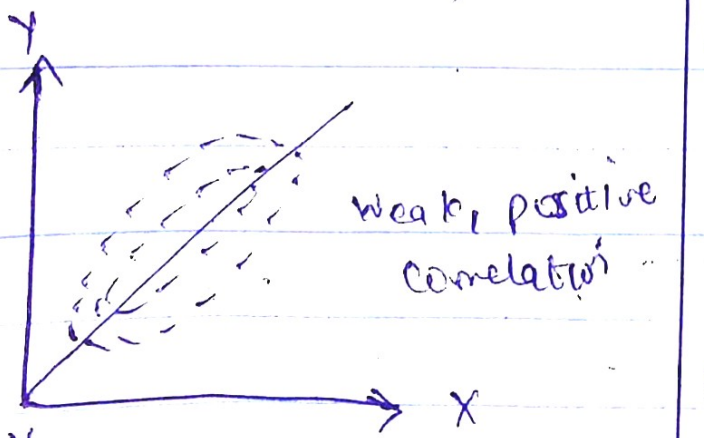


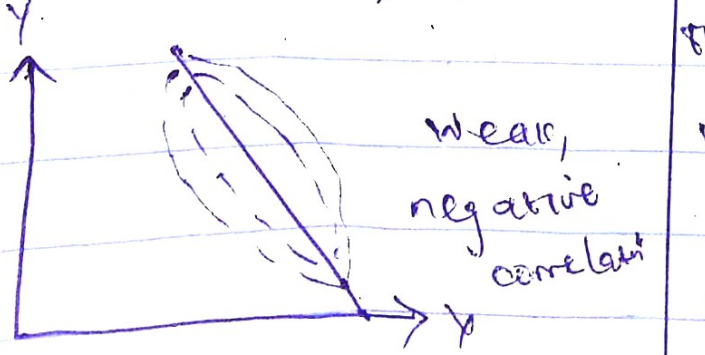
Perfect positive correlation



Zero correlation







The product moment coefficient of correlation (Spearman's Rank coefficient of correlation)

The values of the product moment coefficient of correlation varies between the limits ± 1. A positive sign indicates positive correlation while a negative sign shows negative correlation.

When $r = +1$ the correlation is perfect and positive ~~when r=~~ the correlation is perfect and negative when r approaches $+1$, we have a strong positive correlation when r approaches negative 1

$$\int_{1/2}^{3/2}\left[\int_{1/2}^{2}(6-x-y)\,dx\right]dy$$

$$\int_{1/2}^{3/2}\left[6x-\frac{x^2}{2}-xy\right]_{1/2}^{2}dy$$

when $x=2$ and $=1/2$

$$\left[6(2)-\frac{(2)^2}{2}-2y\right)-\left(6\left(\frac{1}{2}\right)-\frac{\left(\frac{1}{2}\right)^2}{2}-\left(\frac{1}{2}\right)y\right]$$

$$\left[12-2-2y\right)-\left(3-\frac{1/4}{2}-\frac{1}{2}y\right)$$

$$(10-2y)-\left(3-\frac{1}{8}-\frac{1}{2}y\right)dy$$

$$(10-2y)-\left(\frac{24-1-4y}{8}\right)dy$$

$$(10-2y)-\left(\frac{23-4y}{8}\right)dy$$

$$\int_{1/2}^{3/2}\frac{1}{8}\left[\frac{10-2y}{1}-\frac{23+4y}{8}\right)dy$$

$$\frac{1}{8}\left(\frac{80-16y-23+4y}{8}\right)dy$$

$$\int_{1/2}^{3/2}\frac{1}{8}\left(57-12y^2\right)dy$$

$$\frac{1}{8}\times\frac{1}{8}\left[57y-\frac{12y^2}{2}\right]_{2}^{3/2}$$

$$\frac{1}{64}\left[57y-\frac{12y^2}{2}\right]_{2}^{3/2}$$

$$\frac{1}{64}\left[\left(57\times\frac{7}{2}-6\left(\frac{7}{2}\right)^2\right)-57\times2-6(2)^2\right]$$

$$\frac{1}{64}\left(\left(\frac{399}{2}-\frac{147}{2}\right)-(114-24)\right)$$

$$\frac{1}{64}\left(\frac{399-147}{2}-(90)\right)$$

$$\frac{1}{64}\left(\frac{252}{2}-\frac{90}{1}\right)$$

$$\frac{1}{64}(126-90)$$

$$\frac{1}{64}(36)=\frac{18}{32}\;\frac{9}{16}\Rightarrow\frac{9}{16}$$

In exam you can be given

$$f(x) = \begin{cases} \frac{1}{25} x e^{-x/k} & x > 0 \\ 0 & \text{otherwise} \end{cases}$$

Solve later.

Note: $\frac{1}{16} x e^{-x/4}$

$\frac{1}{36} x e^{-x/6}$

$\frac{1}{k} x e^{-x/3}$ ie $k = 9$

$E[ax + b]$
$= a E(x) + b$

Correlation Analysis

While regression analysis discovers the nature of the association between related variables, correlation analysis is concerned with measuring the degree of strength of the relationship

personal attic

Correlation Formula

$$r = \frac{n \Sigma xy - (\Sigma x)(\Sigma y)}{\sqrt{n \Sigma x^2 - (\Sigma x)^2 \cdot n \Sigma y^2 - (\Sigma y)^2}}$$

Regression $\hat{y} = \beta_0 + \beta x$

$$\beta_1 = \frac{n \Sigma xy - \Sigma x \Sigma y}{n \Sigma x^2 - (\Sigma x)^2}$$

Types of correlation

Correlation may be simple or multiple. The degree of the relationship between two variables is called simple correlation. The degree of relationship connecting three or more variables is called multiple correlation. Correlation could be linear or Non-linear. We have a linear correlation when all the pairs of of (x, y) values plotted on a scattered